

Graph Theory

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Introduction

A graph G is a collection of vertices, V and edge, E.

So we say G = <V, E>.

Vertices = dots, Edges = lines

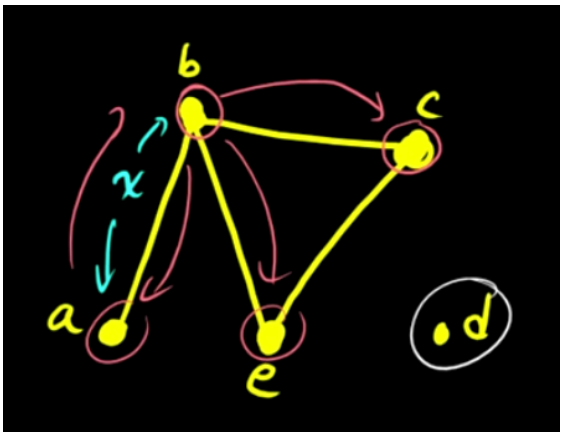
Terminology

Incident: x incident to a and b. Any edge is **incident** to 2 vertices.

Adjacent: b adjacent to a, c and e, vertices that has **edges connecting** is known to be adjacent.

a is only adjacent to b.

Isolated: d is isolated, any vertex without any edges.



Undirected Graphs

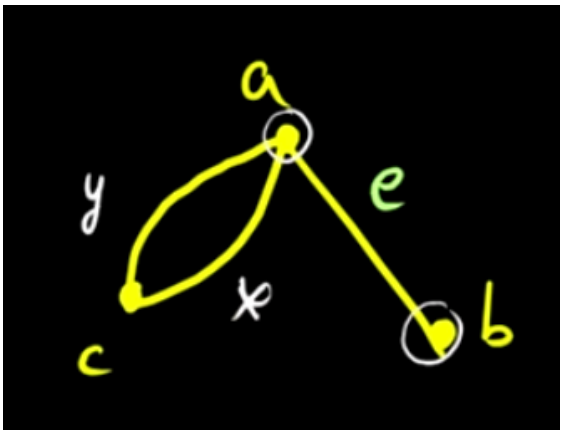
e = {a, b}, we list every edge as a set of vertices. Order does not matter.

y = {a, c}

x = {a, c},

the multiplicity: 2 for {a, c}.

e = ab = ba



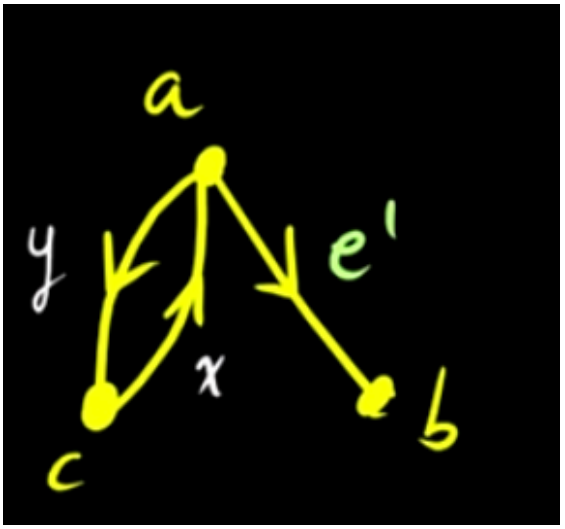
Directed Graphs

e' = {a, b}, order does matter in directed graphs e' = {tail, head}

x = (c, a)

y = (a, c)

e' = ab ≠ ba



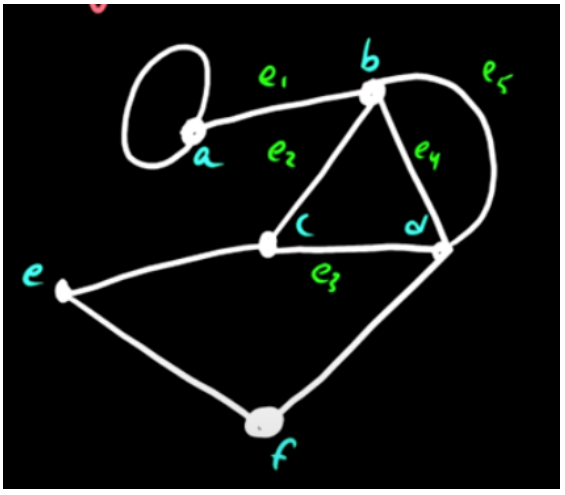
Walk

An x-y walk in G is a sequence of vertex and edge and vertex…

w = x e1 v1 e2 v2 e3 v3 … v_n-1 e_n y,

e1 = x v1, e2 = v1 v2, e3 = v2 v3…

where e_i = {x_{i-1}, x_j}



a → d walk,

a e1 b e5 d

or

a e1 b e2 c e3 d

or

a e1 b e4 d

or *

a e1 b e1 a e1 b e4 d

it is fine to go back and forth in walks

A closed walk occurs when x = y in w.

e → e

e c d f e

or

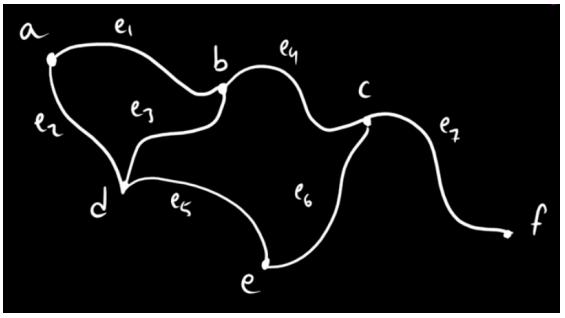
e c e

or

e. (trivial walk it does not take any edges)

Trail

A trail is a walk with no repeated edges.



a d b c f (that is a trail)

e c e d (that is not a trail)

A closed trail is a circuit.

Circuit = walk with no repeated edges where the tail and head is the same

a d b a (is a circuit)

a d a(is not a circuit)

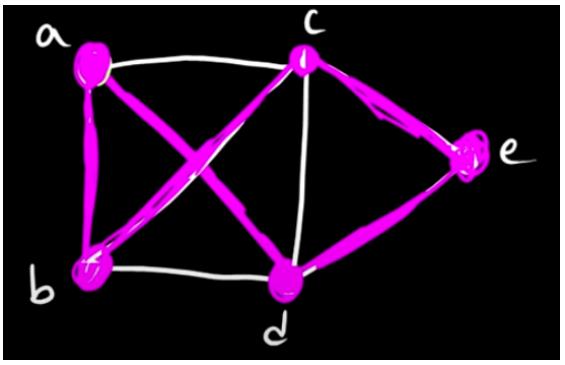
a d e c b a(is a circuit)

Path

A path is a walk with no repeated vertices

A closed path is a cycle

Cycle = path that starts and ends on the same point / edge



a → e

a b d c e (this is a path)

a c b d c e (this is not path)

a → a

a b c a (this is a cycle)

(a is repeated but its in the first and last pos)

a d e c b a (this is a cycle)

Connected Graphs

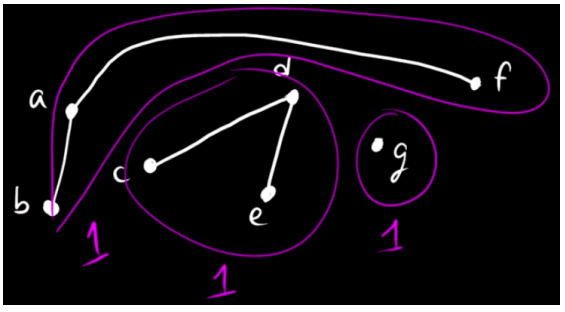
A graph is connected if there is an x-y path for all x, y in V, V = set of vertices.

If every vertex is connected to every another vertex then the graph is connected.

If a graph is not connected, we can take a look at the number of components its have.

number of Components = K(G). (number of connected subgraphs)

Question: Why does a graph have to be an x-y path?

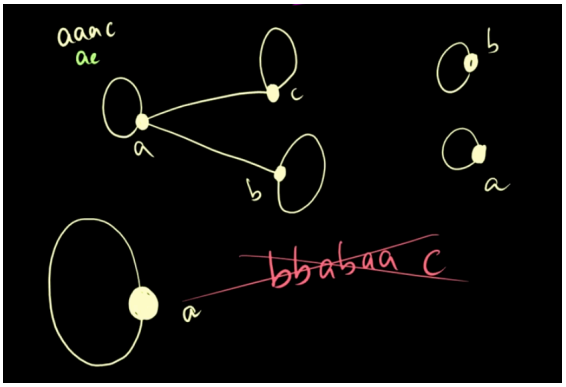


K(G) = 3

Proof:

For All x-y walks, there exist an x-y path

(hint: proof by contradiction, let w = minimal walk from x to y) i.e. w is the



the self loop of a is “pointless”

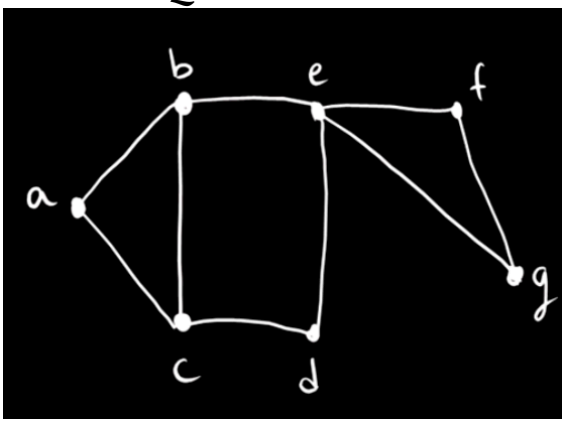
Proof by contradiction: →

shortest walk from x to y

Case 1, no repeated vertex: if there is no repeated vertices, then case 1 is a path by definition. We are given that w is an element of an x, y walk. Since there are no repeated vertices in this walk, x, y is also an element of x-y path.

Case 2, repeated vertex, vi: if there is repeated vertex then that walk is not the shortest path(minimal path) between that x, y pair. We can begin the walk at the repeated vertex to get a shorter walk since that walk will still include the x, y pair as the start and end points. Then we end up with a walk with no repeated vertex and can apply Case 1.

Practice Qs



1) Walk b →d not trail.

trail has repeated edges.

want repeated edges

b e b c d

2) Trail b → d not path

trail has repeat edges

path has no repeated vertices

want repeated vertices

b a c b e d

3) Path b → d

path has no repeated edges

want no repeated edges

b c d (usually the shortest way is a path)

4) Circuit b → b, not cycle

a closed path is a cycle

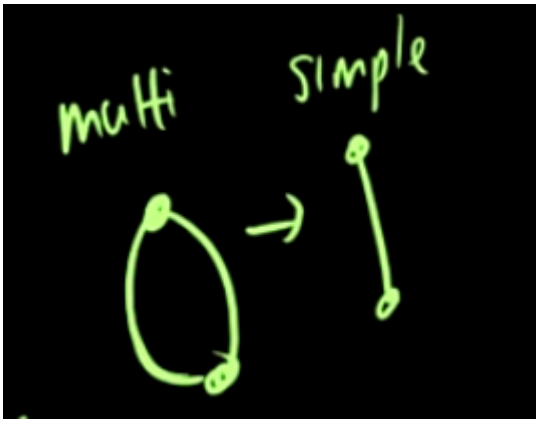
a closed trail is a circuit

never use same edge twice = circuit = don't use a line twice

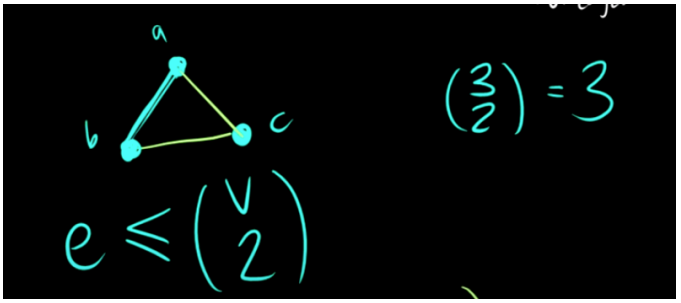
but did use same vertex e = not a cycle = don't use a point twice

b a c d e g f e b

If G = <V, E> where |v| = v(number of vertices) and |E| = e(number of edges) and G is simple (loop free, undirected and no multiple edges), prove that 2e ≤ v^2 - v



Suppose we have 3 vertices, a, b, c.



we can consider edge 1= e1 = a b. How many edges can we generate from the 3 vertices?

(number of edges = e) ≤ (V choose 2)

e ≤ (v(v - 1))/2

2e ≤ (v(v-1))

2e ≤ v^2 - v

(why choose 2?)

because in a simple graph, we need 2 vertices to make an edge, so the number of ways by which we can get a pair of vertices is equal to the maximum number of edges we can form with those vertices!

The way of choosing 2 vertices [pair of vertices] from N number of vertices is N choose 2, so the maximum number of edges we can get is N Choose 2. From there you have the Number of edges ≤ N choose 2.

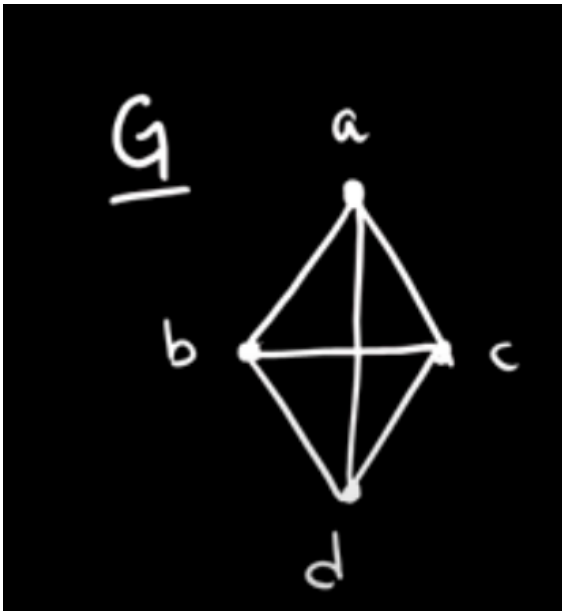
Subgraphs and other…

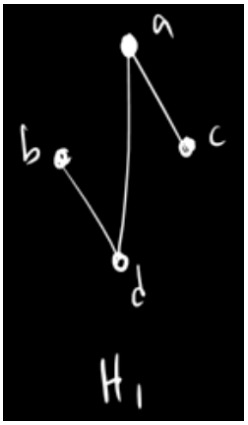
Subgraph

A subgraph H = <Vh, Eh> of G = <Vg, Eg> requires that:

- Vertex set of H is in the Vertex set of G, i.e. Vh is a subset of Vg. (we don't add vertices)
- Set of Edges H is a subset of Set of Edges G, where each e is an element of Eh are incident to v in Vh. (all of the edges in H are incident to the vertices in V)

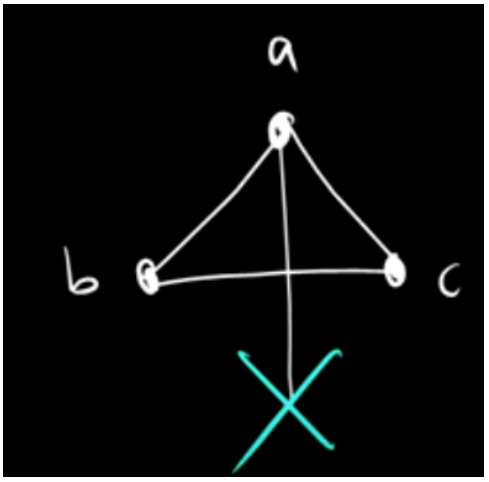
Consider the graph G:





H1 would be a subgraph of G (where we might have deleted (gotten rid off) some edges)

The below is not a subgraph



We would need d to be in where the Big X is.

H4 →

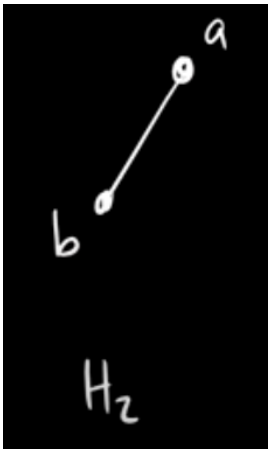
is not a subgraph of G,

because:

e is in H4,

but e is not in G

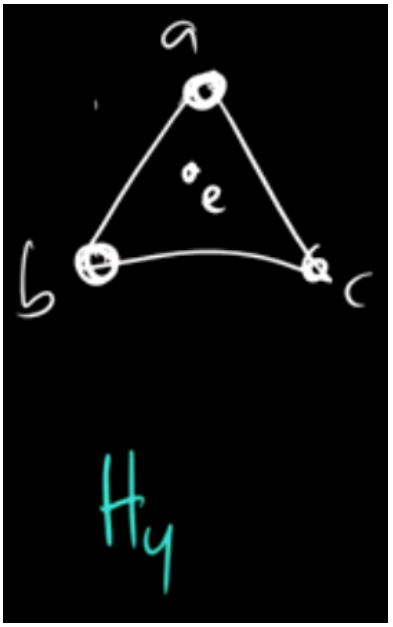




H2 is also a subgraph of G

where a, b is a subset of a, b, c, d

and ab is a subset of ab, ac, ab, ad…

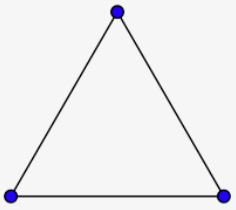
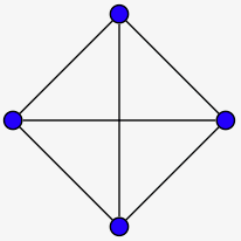
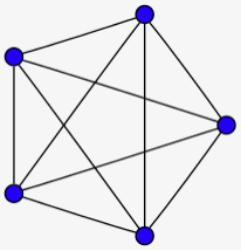
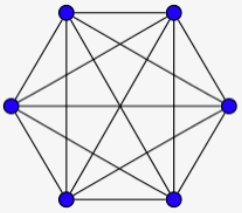
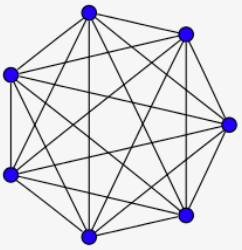
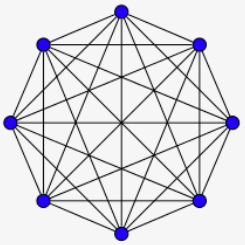


Spanning Subgraph

A spanning subgraph H requires that Vh = Vg.

Complete Graphs

Complete graphs on n vertices, for n between 1 and 12, are shown below along with the numbers of edges:

K_1: 0	K_2: 1	K_3: 3	K_4: 6
			
K_5: 10	**K_6: 15**	**K_7: 21**	**K_8: 28**
			
K_9: 36	**K_{10}: 45**	**K_{11}: 55**	**K_{12}: 66**
